工具-单目相机标定ROS2

准备工作

1.安装依赖：

```
代码块
1   sudo apt install ros-humble-usb-cam
2   sudo apt install ros-humble-camera-calibration
3   sudo apt install ros-humble-cv-bridge
4   sudo apt install ros-humble-image-transport
```

2.准备棋盘格

一个已知尺寸的大棋盘，相机标定使用的是棋盘格内部的相交的顶点个数，本次教程使用的是12×8、2.5cm的棋盘，因此12×8的棋盘对应的顶点个数是11×7

启动相机

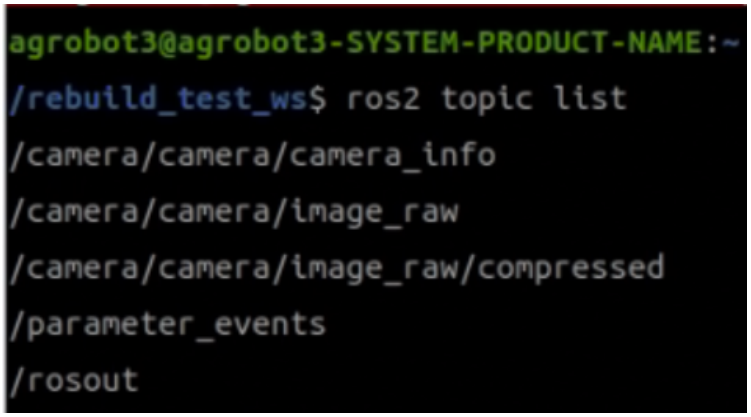
按照相机驱动启动相机

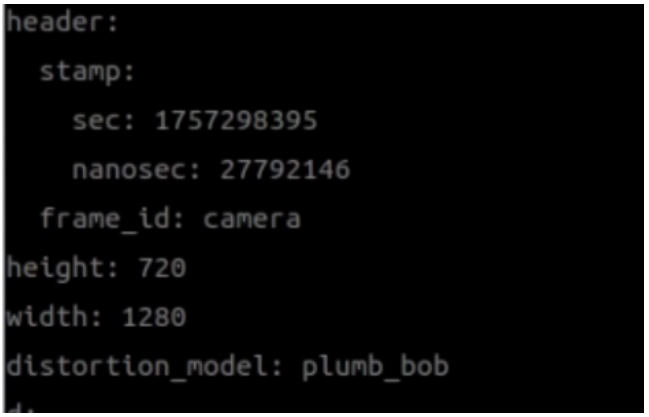
示例：

```
代码块
1   ros2 launch apriltag_ros camera.launch.py
```

查看camera_info

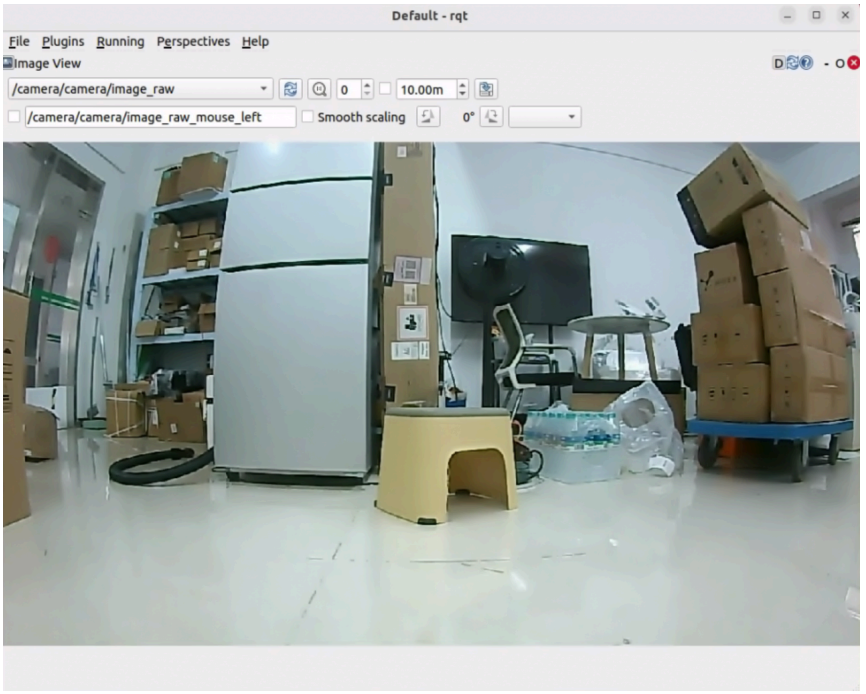
`ros2 topic list`





```
ros2 topic echo /camera/camera/camera_info
```

启动rqt_image_view查看视频流

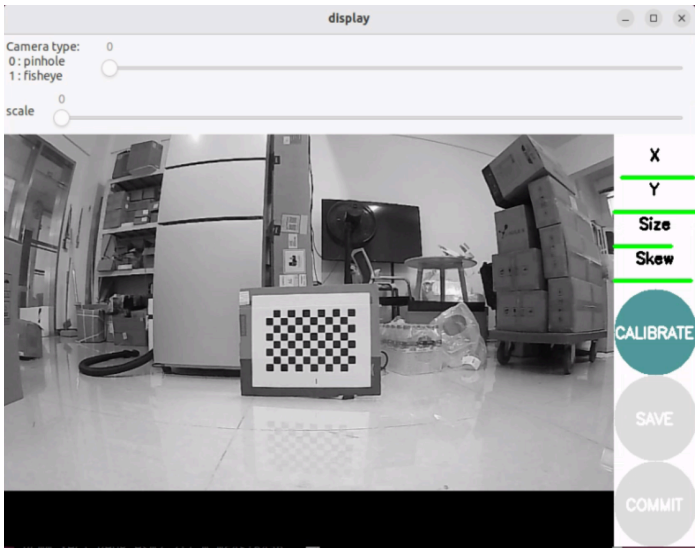


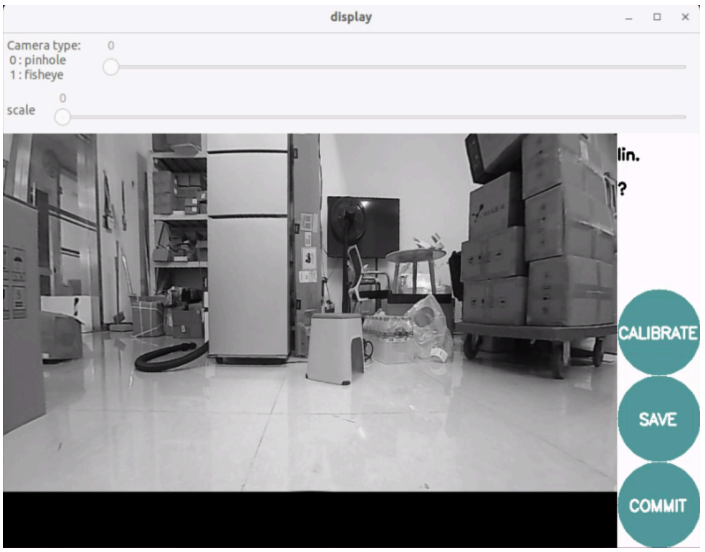
启动相机标定节点

代码块

```
1   ros2 run camera_calibration cameracalibrator --size 11x7 --square 0.025 \
2     --ros-args --remap image:=/camera/camera/image_raw --remap camera:=/camera
```

启动成功后上下左右移动棋盘、前后移动、上下左右倾斜，当X、Y、Size、Skew都变成绿色以后，点击CALIBRATE，等待一分钟左右，中途出现无响应的警告可以忽略，标定完成后点击SAVE，标定文件会保存到/tmp/calibrationdata.tar.gz





文件中的ost.yaml即为标定后的参数文件，应用到相机启动launch即可